$(t_0\ \sigma_0\ Q\ ((\text{stack}\ (\text{frame}\ E[((\text{async/lambda}\ (x\ \ldots\_1)\ e)\ v\ \ldots\_1)]\ l)\ F\ \ldots)))\ \longrightarrow$ [async-app]
$(t_1\ \sigma_1\ Q\ ((\text{stack}\ (\text{frame}\ e\ x_{async})\ (\text{frame}\ E[(\text{task}\ x_{async})]\ l)\ F\ \ldots)))$

$(t_0\ \sigma_0\ Q_0\ ((\text{stack}\ (\text{frame}\ v\ l)\ F\ \ldots)))\ \longrightarrow$ [task-return]
$(t_1\ \sigma_1\ Q_1\ ((\text{stack}\ F\ \ldots)))$

where $(\texttt{async?}\ l)$, $x_{async} = l$, $v_{obj} = \text{lookup}[\![\sigma_0, x_{async}]\!]$

$(t_0\ \sigma\ Q_0\ ((\text{stack}\ (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)\ F\ \ldots)))\ \longrightarrow$ [await-continue]
$(t_1\ \sigma\ Q_1\ ((\text{stack}\ F\ \ldots)))$

where $(\text{done}\ v) = \text{task-status}[\![\text{lookup}[\![\sigma, x_{async}]\!]]\!]$

$(t_0\ \sigma\ Q_0\ ((\text{stack}\ (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)\ F\ \ldots)))\ \longrightarrow$ [await-failed]
$(t_1\ \sigma\ Q_1\ ((\text{stack}\ F\ \ldots)))$

where $(\text{failed}\ v) = \text{task-status}[\![\text{lookup}[\![\sigma, x_{async}]\!]]\!]$

$(t_0\ \sigma_0\ Q\ ((\text{stack}\ \text{current-frame}\ F\ \ldots)))\ \longrightarrow$ [await]
$(t_1\ \sigma_1\ Q\ ((\text{stack}\ F\ \ldots)))$

where $v_{obj} = \text{lookup}[\![\sigma_0, x_{async}]\!]$, $(\text{pending}\ \_\ \ldots) = \text{task-status}[\![v_{obj}]\!]$, $\text{current-frame} = (\text{frame}\ E[(\text{await}\ (\text{task}\ x_{async}))]\ l)$

$(t_0\ \sigma_0\ Q_0\ ((\text{stack}\ (\text{frame}\ E[(\text{throw}\ v_{err})]\ l)\ F\ \ldots)))\ \longrightarrow$ [async-throw]
$(t_1\ \sigma_1\ Q_1\ ((\text{stack}\ F\ \ldots)))$

where $(\texttt{async?}\ l)$, $(\texttt{not}\ \text{in-handler?/js}[\![E]\!])$, $x_{async} = l$, $v_{obj} = \text{lookup}[\![\sigma_0, x_{async}]\!]$

$(t_0\ \sigma_0\ Q_0\ ((\text{stack}\ (\text{frame}\ E[(\text{os/io}\ \textit{natural}\ v)]\ l)\ F\ \ldots)))\ \longrightarrow$ [os/io]
$(t_1\ \sigma_1\ Q_1\ ((\text{stack}\ (\text{frame}\ E[(\text{task}\ x_{async})]\ l)\ F\ \ldots)))$

where $(\text{ptr}\ x_{async}) = \text{malloc}[\![\sigma_0]\!]$

$(t_0\ \sigma_0\ Q_0\ ((\text{stack}\ (\text{frame}\ E[(\text{os/resolve}\ (\text{task}\ x_{async})\ t_{resolve}\ v)]\ l)\ F\ \ldots)))\ \longrightarrow$ [os/resolve]
$(t_1\ \sigma_1\ Q_1\ ((\text{stack}\ F\ \ldots)))$

where $(>=\ t_0\ t_{resolve})$, $v_{obj} = \text{lookup}[\![\sigma_0, x_{async}]\!]$

$(t_0\ \sigma\ Q_0\ ((\text{stack}\ \text{current-frame}\ F\ \ldots)))\ \longrightarrow$ [os/resolve-blocking]
$(t_1\ \sigma\ Q_1\ ((\text{stack}\ F\ \ldots)))$

where $(<\ t_0\ t_{resolve})$, $\text{current-frame} = (\text{frame}\ E[(\text{os/resolve}\ v_{task}\ t_{resolve}\ v)]\ l)$

$(t_0\ \sigma\ Q_0\ ((\text{stack})))\ \longrightarrow$ [thread-work-steal]
$(t_1\ \sigma\ Q_1\ ((\text{stack}\ F_{head})))$

where $(F_{head}\ Q_1) = \text{q-pop}[\![Q_0]\!]$

$(t_0\ \sigma_0\ Q\ ((\text{stack}\ (\text{frame}\ e_0\ l)\ F\ \ldots)))\ \longrightarrow$ [base-step]
$(t_1\ \sigma_1\ Q\ ((\text{stack}\ (\text{frame}\ e_1\ l)\ F\ \ldots)))$

where $(\texttt{not}\ \text{value?}[\![e]\!])$, $(\sigma_1\ e_1) = \Downarrow\text{base}[\![\sigma_0, e_0]\!]$

$(t_0\ \sigma\ Q_0\ ((\text{stack}\ \text{current-frame}\ )))\ \longrightarrow$ [block]
$(t_1\ \sigma\ Q_1\ ((\text{stack})))$

where $(\texttt{sync?}\ l)$, $(\text{pending}\ \_\ \ldots) = \text{task-status}[\![\text{lookup}[\![\sigma, x_{async}]\!]]\!]$, $\text{current-frame} = (\text{frame}\ E[(\text{block}\ (\text{task}\ x_{async}))]\ l)$

$(t_0\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{block}\ (\text{task}\ x_{async}))]\ l))))\ \longrightarrow$ [unblock]
$(t_1\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[v]\ l))))$

where $(\texttt{sync?}\ l)$, $(\text{done}\ v) = \text{task-status}[\![\text{lookup}[\![\sigma, x_{async}]\!]]\!]$

$(t_0\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{block}\ (\text{task}\ x_{async}))]\ l))))\ \longrightarrow$ [unblock-failed]
$(t_1\ \sigma\ Q\ ((\text{stack}\ (\text{frame}\ E[(\text{throw}\ v)]\ l))))$

where $(\texttt{sync?}\ l)$, $(\text{failed}\ v) = \text{task-status}[\![\text{lookup}[\![\sigma, x_{async}]\!]]\!]$